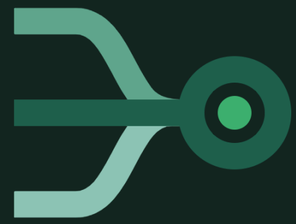
TRACK · BEST APPS & AGENTS

NEBIUS × NVIDIA GLOBAL AI HACKATHON 2026

A regulated decision where *AI reads*, deterministic code writes the reply, and a person decides.

Verdikt reads inbound case files, decides against a program's own human-authored rules with citations, and issues a reply letter that no language model touched.



Verdikt.

AI-in, deterministic-out.
A decision pipeline for regulated document workflows.

THE PROBLEM · PATIENT ASSISTANCE PROGRAM INTAKE

The decision is small. The reading is what takes days.

TODAY · MANUAL

A person reads every page

- Opens the enrollment form, income proof, insurance card and prescription
- Works out which program rule each fact answers
- Passes the file to the next team, who re-reads it

NOW POSSIBLE · AI AGENT

A model reads the packet

- Scans and forms have no fixed layout, and a vision model can read them
- A model can classify the case and pick the rules
- But if it also writes the letter, the audited artifact is probabilistic

THE NEW PROBLEM

Who stands behind the letter?

- Review everything and the gain is lost
- Review nothing and one error reaches a patient
- The letter must be reproducible years later

The product: reads the packet once, decides against the program's own rules, cites the rule that fired.
The pattern: the same split fits any regulated document workflow where the letter is the audit artifact.

THE SOLUTION · ARCHITECTURAL RATIONALE

Draw the line where the *trust requirement* changes.

Reading may be wrong, so it stays reviewable. Writing must be reproducible, so it never touches a model.

01

AI reads

Extraction, triage, retrieval and a structured verdict. Each step has a fallback and an audit entry.

02

Code writes

The reply is a merge of a frozen, human-approved template with the verdict. No model on this path.

03

The line is enforced

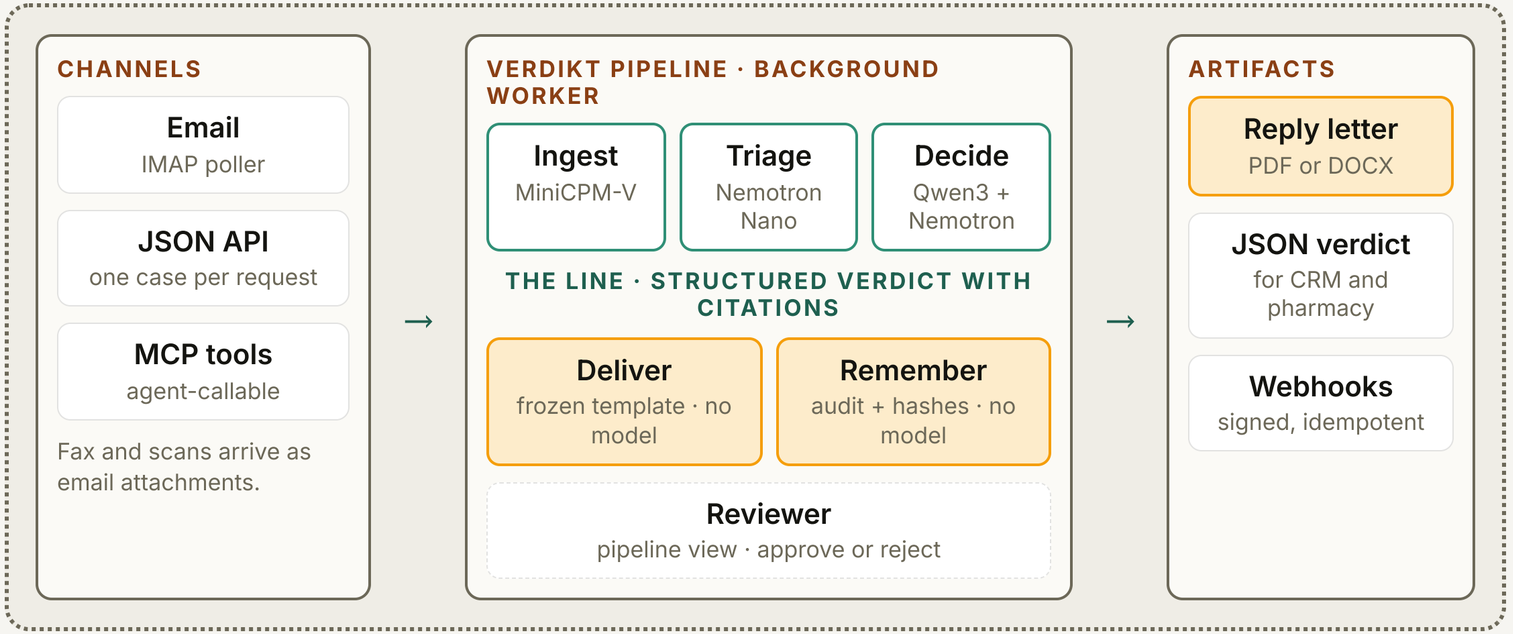
A boundary test fails the build if the render core ever imports the AI layer.

WHAT WE SHOW

A reply letter you can trace to the rule that fired, the template version that produced it, and the stage that handled each step.

ARCHITECTURE · AI-IN, DETERMINISTIC-OUT

Many channels in, one case, two artifacts out.



Channels in

Everything normalizes to one case before the pipeline runs.

AI-in · reviewable

Models on one Nebius Token Factory endpoint. Every stage writes an audit entry.

Deterministic-out

A frozen template merge. Same case, same template version, same reply.

Platform · FastAPI web · background worker · Postgres · Redis · S3-compatible storage · tenant-scoped

HOW IT WORKS

Three steps, one endpoint.

① READ · INGEST AND TRIAGE

The models read

INGEST

MiniCPM-V reads attachments into the program's own field schema.

TRIAGE

Nemotron Nano classifies the case and routes it. A rule-based classifier is the fallback.

② DECIDE · RETRIEVE AND REASON

The rules decide

- Qwen3-Embedding retrieves the rules that apply to this case
- Nemotron Super returns approve, deny or missing info
- Every verdict cites the rule that fired

③ DELIVER · REMEMBER

The template writes

- A frozen, human-approved template, merged with the verdict
- No model on the render path
- One audit entry per stage, with template hash and render fingerprint

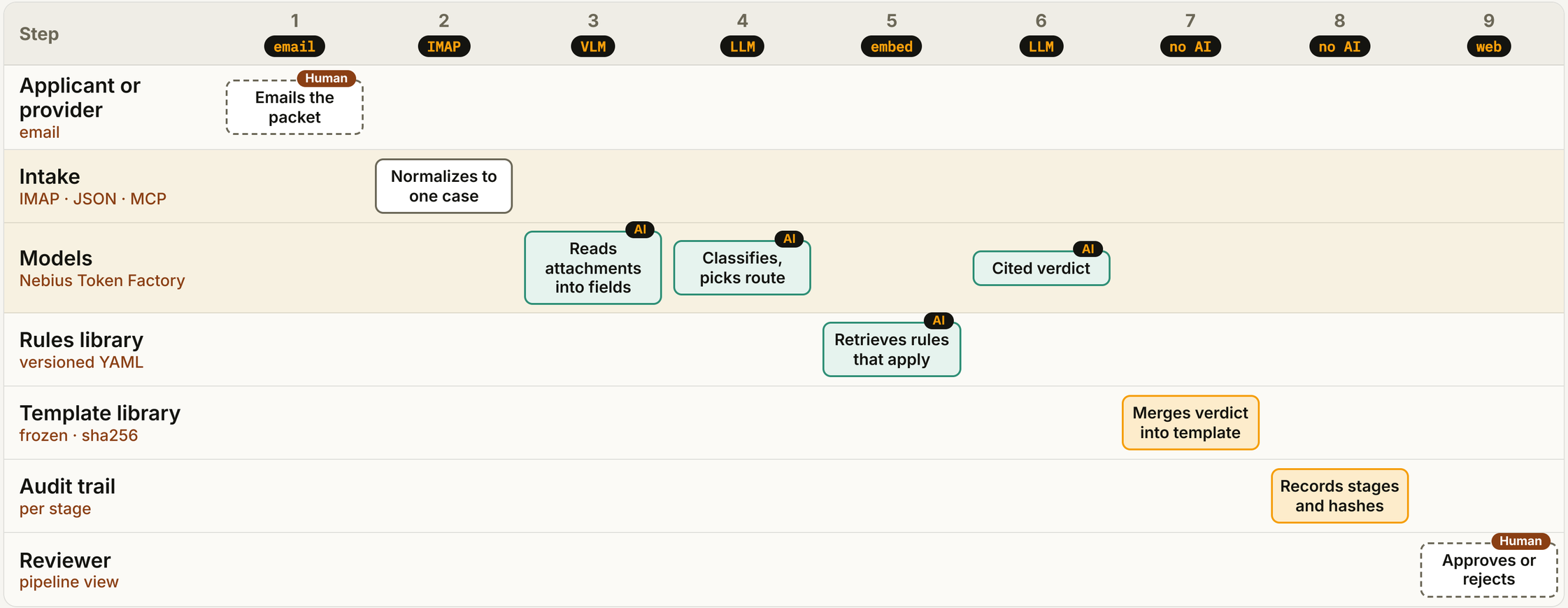
Models run on one Nebius Token Factory endpoint.

Rules are human-authored and versioned.

Templates are frozen before they can render.

BUSINESS FLOW · HAPPY PATH

One packet in, one sealed reply out.



Solid: every case. **Dashed:** a person acts. **AI:** a model's judgment. **No AI:** deterministic code.

TRIAGE AND PROGRAMS AS DATA

Not every case needs a verdict.

CASE 1 · NEW INTAKE

Decide, then deliver

The full path: retrieve rules, cited verdict, frozen template.

CASE 2 · REFILL

Straight to pharmacy

No verdict needed, so Decide is skipped and the audit row stays.

CASE 3 · NOT DECIDABLE

Rejected, on the record

An audit row and no further action. Appeals go to the CRM queue.

THE CLOSER · SAME HOUSEHOLD, TWO PROGRAMS

Cardiovita: cap 400% of the poverty guideline = $84,600. A $60,000 household passes.

OncoBridge: cap 250% = $52,875. The same household fails. Same code, different data.

FAILS SAFE

If the triage model fails or is unsure, a rule-based classifier takes over, so triage never blocks the pipeline.

ROUTES ARE DATA

The routing table is overridable per program or tenant. A change is a row, not a deploy.

 Demo data is fictional

EVIDENCE

Evidence, not adjectives.

921	916	0	5
tests in the suite	passing	failing	skipped

THE SPLIT IS ENFORCED

Boundary tests fail the build if the render core imports the AI layer.

THE LABELER MEASURABLY IMPROVED

On one dense 104-field form, correct labels went from 24 to 98 after moving to vision with chunk and crop.

EVERY STAGE IS ON THE RECORD

One audit entry per stage, with the model that handled it and the time it took.

 Scope: test counts from the latest full run; the labeler result is one benchmark

MODELS · NEBIUS TOKEN FACTORY

Four models, *one endpoint*.

STAGE	MODEL	JOB
Ingest	openbmb/MiniCPM-V-4_5	Multimodal extraction
Triage	NVIDIA Nemotron Nano	Case classification and routing
Decide	Qwen/Qwen3-Embedding-8B	Rule retrieval
Decide	NVIDIA Nemotron Super 120B	Cited, structured verdict

ONE ENDPOINT, ONE KEY

Every model call goes through Nebius Token Factory.

SWAPPABLE BY CONFIGURATION

The model for each stage is set per tenant or program, with no code change.

NO MODEL AT RENDER

The reply letter comes from a frozen template, never from a model.

BUSINESS VALUE

Where the value comes from, and *how to measure it*.

DEMONSTRATED

- Every verdict cites the rule that fired
- Refills skip the verdict entirely
- The reply is merged, never generated
- A second program runs on the same code

BY DESIGN

- A new program is configuration, not a build
- A new channel is one adapter
- Models swap per program
- Reproducible, auditable replies

TO MEASURE IN A PILOT

- [__]% of cases a reviewer re-opens the source PDF
- [__] median time from receipt to verdict
- [__] cost per case

The pattern generalizes: prior authorization, benefit verification, informed consent and safety reporting share the same shape. A regulated letter that is also the audit artifact.

 Bracketed figures: to be filled from a measured pilot

WHAT SHIPS NEXT

Honest about *what is left*.

01

Deploy on Nebius AI Cloud

The runbook and checklist are written; the models already run on Token Factory.

02

Byte-level reproducibility

Replay a stored case in CI and assert the letter's bytes, not only its structure.

03

A measured pilot

Fill the "to measure" figures from real cases.

THE ONE-LINE VERSION

AI reads. Deterministic code writes. Humans decide.